File

Name: P4-fit-A3
Description: KISSsoft example
Changed by: AAA on: 30.10.2026 at: 19:34:49

1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00	/10.00
Diameter hub (mm)	[DaA]/[DaI]	125.00	/50.00
Length of Interference fit (mm)	[l]	40.00	
Diameter of joint (mm)	[DF]	50.00	
with special allowances:			
Upper allowance Shaft (µm)	[AoI]	73.9	
Lower allowance Shaft (µm)	[AuI]	73.9	
Tolerance measure Shaft (µm)	[TI]	0.0	
Upper allowance Hub (µm)	[AoA]	0.0	
Lower allowance Hub (µm)	[AuA]	0.0	
Tolerance measure Hub (µm)	[TA]	0.0	
Largest interference (µm)	[Po]	73.9	
Smallest interference (µm)	[Pu]	73.9	
Nominal torque (Nm)	[T]	0.00	
Application factor	[KA]	1.00	
Service torque (Nm)	[Tb]	0.00	
Axial force (N)	[FA]	0.00	
Bending moment (Nm)	[Mb]	0.00	
Radial force (N)	[Fr]	0.00	
Circumferential force (N)	[Fu]	0.00	
Speed (1/min)	[n]	0.00	
Interference (µm)	[P]	73.9	(73.9 .. 73.9)
Embedding (µm)	[s]	3.84	
Effective interference at 20°C(68°F) (µm)	[Pw]	70.10	(70.1 .. 70.1)
Effective interference at working temp.. (µm)	[PwTh]	70.10	(70.1 .. 70.1)
Service temperature shaft (°C)	[ThB]	20.0	
Service temperature hub (°C)	[ThB]	20.0	
Pressure stress by			
- Interference (after mounting) (N/mm²)	[pM]	50.69	(50.69 .. 50.69)
- Interference (working) (N/mm²)	[p]	50.69	(50.69 .. 50.69)
- Bending moment (N/mm²)	[pb]	0.00	
- Radial force (N/mm²)	[pr]	0.00	
Coefficient of friction, longitudinal	[µa]	0.120	
Coefficient of friction, circumferential	[µu]	0.120	
Safety against sliding	[Sr]	> 1000	
Required safety against sliding	[SSr]	1.50	

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	650.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The shaft's material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	21			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-9.23	(-9.23	..	-9.23)
Diameter increase during operation (o/oo)	[ΔD]	-0.18	(-0.18	..	-0.18)
Equivalent stress outer diameter (N/mm²)	[σVa]	52.93	(52.93	..	52.93)
- Radial stress (N/mm²)	[σRa]	-50.69	(-50.69	..	-50.69)
- Tangential stress (N/mm²)	[σTa]	-54.92	(-54.92	..	-54.92)
- With outside load (N/mm²)	[σVaMF]	52.93	(52.93	..	52.93)
Inner diameter (mm)	[Dil]	10.00			
Diameter increase during operation (µm)	[ΔD]	-4.91	(-4.91	..	-4.91)
Diameter increase during operation (o/oo)	[ΔD]	-0.49	(-0.49	..	-0.49)
Equivalent stress inner diameter (N/mm²)	[σVi]	105.61	(105.61	..	105.61)
- Radial stress (N/mm²)	[σRi]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTi]	-105.61	(-105.61	..	-105.61)
Safety against fracture	[SiRm]	6.15	(6.15	..	6.15)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	2.84	(2.84	..	2.84)
Required safety against yield point	[SSi.Rp]	1.20			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	70000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	150.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	40			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DaA]	125.00			
Diameter increase during operation (µm)	[ΔD]	34.49	(34.49	..	34.49)
Diameter increase during operation (o/oo)	[ΔD]	0.28	(0.28	..	0.28)
Equivalent stress outer diameter (N/mm²)	[σVa]	19.31	(19.31	..	19.31)
- Radial stress (N/mm²)	[σRa]	0.00	(0.00	..	0.00)
- Tangential stress (N/mm²)	[σTa]	19.31	(19.31	..	19.31)
Inner diameter (mm)	[Dal]	50.00			
Diameter increase during operation (µm)	[ΔD]	60.87	(60.87	..	60.87)
Diameter increase during operation (o/oo)	[ΔD]	1.22	(1.22	..	1.22)
Equivalent stress inner diameter (N/mm²)	[σVi]	104.97	(104.97	..	104.97)
- Radial stress (N/mm²)	[σRi]	-50.69	(-50.69	..	-50.69)
- Tangential stress (N/mm²)	[σTi]	70.01	(70.01	..	70.01)

- Radial stress with outside load (N/mm²)	[σRai]	-50.69	(-50.69	..	-50.69)
- Tangential stress with outside load (N/mm²)	[σTai]	70.01	(70.01	..	70.01)
- With outside load (N/mm²)	[σViMF]	104.97	(104.97	..	104.97)
Safety against fracture	[Si.Rm]	6.67	(6.67	..	6.67)
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Re]	1.43	(1.43	..	1.43)
Required safety against yield point	[SSi.Re]	1.20			

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:					
Mounting clearance (mm)	[PsTh]	0.050			
Temperature difference for mounting:					
Shaft temperature: (°C) Hub temperature: [ThA] (°C)					
Mount with shaft temperature 20 °C:					
Assembly temperature shaft (°C)	[θI]	20			
Assembly temperature hub (°C)	[θA]	235.552			
Mount with shaft temperature -150 °C:					
Assembly temperature shaft (°C)	[θI]	-150			
Assembly temperature hub (°C)	[θA]	109.900			
calculated using the coefficient of thermal expansion					
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50			
Longitudinal pressure fit:					
Assembly temperature shaft (°C)	[θM]	20.00			
Assembly temperature hub (°C)	[θM]	20.00			
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16		
Press on (force) (kN)	[Fpress]	49.69	(49.69	..	49.69)
Coefficient of friction, longitudinal	[μll=μa*1.600]	0.19			
Press out (force) (kN)	[Fpress]	61.16	(61.16	..	61.16)
Coefficient of friction	[μ]	0.19			
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	309.61	(309.61	..	309.61)

Notice concerning the display: Number-1 (Number-2.. Number-3):

Number-1: Value calculated with the mean allowance

Number-2: Value of the smallest possible allowance

Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 157)
